

Quartz resonator — Solution

X26 (2026) — English translation

A0

?? pts

Write down the code for the package containing the electronic components on your answer sheets. Check that it is complete.

To complete this step, you need to carefully look around the entire workplace. If you look carefully enough, there is a high probability of finding a package with electronic components on the table, on which an individual code in the “*OOO 00 00*” format is applied. After discovering the package, we carefully rewrite the entire code using a pen into a specially designated place on the answer sheet (this place is easily identified by the inscription “A0” on the left). At this stage you need to be extremely focused, otherwise it may lead to mistakes. The most common mistakes are: rewriting only the last 4 characters (“*00 00*”); rewriting characters in the wrong order (“*00 00 OOO*”); errors in spelling of characters (“*OOO oo oo*”); adding extra characters (“*00 00 (OOO)*”); no code on the answer sheet. The next step in completing this item is to check the completeness of the package. The check procedure is as follows: Open the bag. Carefully pour out all the ingredients from the bag. Make sure the bag is really empty. Familiarize yourself with the list of equipment from the condition (if you are sufficiently insinuating when reading the condition of the problem, it can be found by the inscription Equipment). Check for the presence of a quartz resonator in the poured elements (looks like a gray box with legs). Read the table of capacitor markings presented after the photo of the equipment in the condition. Attention! It is required to compare in order the presence of capacitors in the table (see point 3) and on the table, poured out of the bag in point 1 (see point 1). If the mapping between the names in the table and the capacitors on the table is one-to-one, then you can proceed to the next step. Otherwise, you should contact the duty officer.

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After discovering the package, we carefully rewrite the entire code using a pen into a specially designated place on the answer sheet (this place is easily identified by the inscription “A0” on the left). At this stage you need to be extremely focused, otherwise it may lead to mistakes. The most common mistakes are:

The next step in completing this item is to check the completeness of the package. The verification procedure is as follows:

Answer: Answer:

OOO 00 00

A1

0.20 pts

Find the signal frequency $f_{\max}$ at which the voltage amplitude across the quartz resonator is maximum. The difference between $f_{\max}$ and 2 MHz is no more than 1 kHz.

Answer: Answer:

$f_{\max} = 1.999860 \text{ MHz.}$

A2**0.05 pts**

Record the ratio of voltage amplitudes on two oscilloscope channels $\frac{U_{CH2}}{U_{CH1}}$ at signal frequency $f_{\max}$.

Answer: Answer:

$$\frac{U_{CH2}}{U_{CH1}} = 0.733.$$

A3**2.80 pts**

Remove the dependence of $2U_{CH2}$ and φ on the frequency f in the range $[f_{\max} - 250 \text{ Hz}; f_{\max} + 250 \text{ Hz}]$, where φ is the phase difference between the signals from channels 2 and 1. Measure at least 30 points from the given frequency range.

Answer: Answer: f , MHz U_{CH2} , mV Δt , ns φ , rad $|Z|$, $\square$ f , MHz U_{CH2} , mV Δt , ns φ , rad $|Z|$, $\square$

1.999610	80.0	72	0.905	852	1.999850	372	0	0.000	12482	1.999630	77.6	72	0.905	
825	1.999855	376	12	0.151	11808	1.999650	72.0	68	0.854	765	1.999858	376	16	0.201
11101	1.999660	64.8	88	1.106	663	1.999860	384	26	0.327	9685	1.999670	58.8	84	1.055
603	1.999865	380	32	0.402	8458	1.999680	55.6	84	1.055	568	1.999870	376	42	0.528
6978	1.999690	52.0	84	1.055	530	1.999875	364	44	0.553	6466	1.999700	49.2	80	1.005
502	1.999880	352	58	0.729	5113	1.999710	44.8	78	0.980	457	1.999890	332	64	0.804
4471	1.999720	41.2	64	0.804	423	1.999900	312	74	0.930	3788	1.999730	36.4	56	0.704
374	1.999910	292	78	0.980	3422	1.999740	34.8	34	0.427	360	1.999920	276	82	1.030
3132	1.999750	35.6	0	0.000	372	1.999930	258	80	1.005	2970	1.999760	39.6	-26	-0.327
415	1.999940	246	84	1.056	2751	1.999770	50.4	-38	-0.477	536	1.999950	236	84	1.056
2637	1.999780	67.2	-50	-0.628	727	1.999960	228	86	1.081	2513	1.999790	88.8	-58	-0.729
981	1.999970	220	86	1.081	2423	1.999800	134	-54	-0.679	1597	1.999980	212	90	1.131
2280	1.999810	158	-56	-0.704	1936	1.999990	208	86	1.081	2287	1.999815	170	-52	-0.653
2154	2.000000	202	88	1.106	2194	1.999820	206	-54	-0.679	2722	2.000010	200	88	1.106
2172	1.999825	222	-44	-0.553	3191	2.000020	196	92	1.156	2083	1.999830	266	-49	-0.616
3953	2.000030	192	92	1.156	2040	1.999835	284	-32	-0.402	5079	2.000040	190	92	1.156
2019	1.999838	302	-22	-0.276	6221	2.000050	188	92	1.156	1997	1.999840	322	-20	-0.251
7243	2.000070	182	80	1.005	2048	1.999843	332	-14	-0.176	8239	2.000090	176	76	0.955
2012	1.999845	342	-10	-0.126	9199	2.000110	174	80	1.005	1950	1.999848	356	-6	-0.075
10610														

A4**2.50 pts**

Plot a graph of the impedance modulus of the quartz resonator $|Z|$ versus frequency f over the measured range.

Let's introduce notations for the voltages that we measure on an oscilloscope:

$$U_0 \equiv U_{CH1}; \quad U_1 \equiv U_{CH2}.$$

Let's write the equations for amplitudes and phases for two channels of the oscilloscope:

$$\tilde{U}_{CH1} = \tilde{I}(R_0 + Z);$$

$$\tilde{U}_{CH2} = \tilde{I}Z.$$

Divide the equations and get:

$$\frac{U_{CH2}}{U_{CH1}} e^{i\varphi} = \frac{Z}{R_0 + Z}.$$

Now we can get an expression for the impedance of a quartz resonator:

$$Z = U_{CH2} R_0 \cdot \frac{e^{i\varphi}}{U_{CH1} - U_{CH2} e^{i\varphi}}$$

This means that the impedance module can be recalculated using the following formula:

$$|Z| = \frac{U_{CH2} R_0}{\sqrt{(U_{CH1} - U_{CH2} \cos \varphi)^2 + (U_{CH2} \sin \varphi)^2}}$$

Let's recalculate (see solution A3) and plot the dependence.

$$f_0 = 1.999000 \text{ MHz}$$

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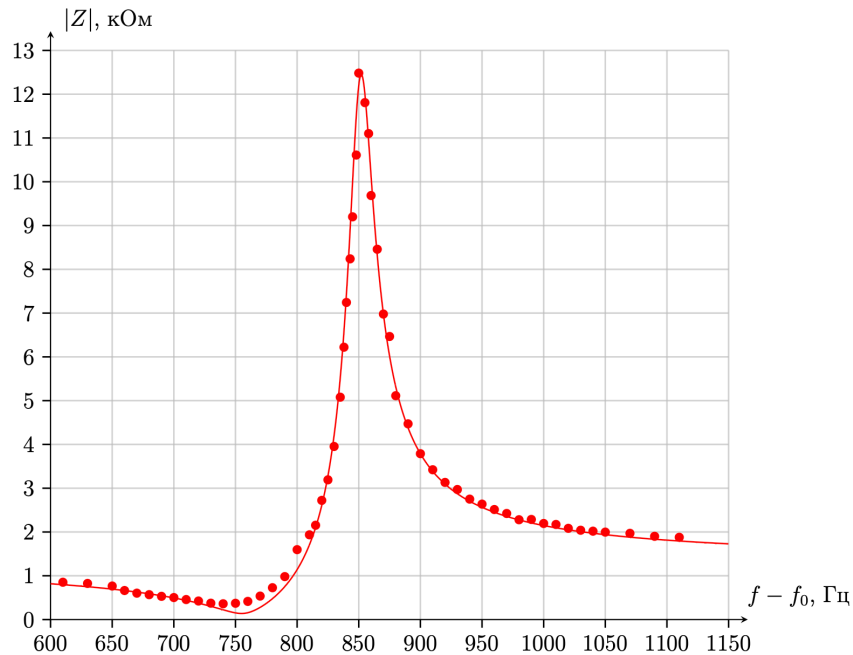
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Let's perform the recalculation (see solution A3) and plot the dependence.

$$f_0 = 1.999000 \text{ MHz}$$



A5

0.10 pts

From the plotted graph $|Z|(f)$, find the frequencies f_{res} and f at which $|Z|$ is maximum and minimum, respectively.

Answer: Answer:

$$f_{res} = 1.999850 \text{ MHz}; \quad f = 1.999740 \text{ MHz}.$$

A6

0.30 pts

Express $\omega_s, \omega_p, \alpha$ in terms of R, L, C, C_0 .

Impedance of parallel connection of capacitor C_0 and RLC -chain:

$$\frac{1}{Z} = \frac{1}{R + i\omega L + \frac{1}{i\omega C}} + i\omega C_0$$

Let's simplify the expression:

$$\frac{1}{Z} = \frac{1 + i\omega C_0 (R + i\omega L + \frac{1}{i\omega C})}{R + i\omega L + \frac{1}{i\omega C}}$$

$$Z(\omega) = \frac{\frac{1}{LC} - \omega^2 + i\omega \frac{R}{L}}{i\omega C_0 \left(\frac{1}{LC_0} + \frac{1}{LC} - \omega^2 + i\omega \frac{R}{L} \right)}$$

Answer: Answer: Thus:

$$\omega_s = \sqrt{\frac{1}{LC}}; \quad \omega_p = \sqrt{\frac{1}{LC} \left(1 + \frac{C}{C_0} \right)}; \quad \alpha = \frac{R}{L}.$$

A7

0.10 pts

Using the approximations described above, express $|Z|^2(\omega)$ in terms of $\alpha, \omega, \omega_s, \omega_p, C_0$.

Let us take advantage of the fact that for any two complex numbers z_1, z_2 $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$ holds:

Answer: Answer:

$$|Z|^2 = \frac{(\omega_s^2 - \omega^2)^2 + (\omega_p \alpha)^2}{(\omega_p C_0)^2 ((\omega_p^2 - \omega^2)^2 + (\omega_p \alpha)^2)}.$$

A8

0.20 pts

Using the approximations described above, express f_{res} and f in terms of ω_s, ω_p .

$$\frac{d|Z|^2}{d\omega^2} = \frac{2}{(\omega_p C_0)^2} \cdot \frac{-(\omega_s^2 - \omega^2)((\omega_p^2 - \omega^2)^2 + (\omega_p \alpha)^2) + (\omega_p^2 - \omega^2)((\omega_s^2 - \omega^2)^2 + (\omega_p \alpha)^2)}{((\omega_p^2 - \omega^2)^2 + (\omega_p \alpha)^2)^2}$$

$$\frac{d|Z|^2}{d\omega^2} = \frac{2}{(\omega_p C_0)^2} \frac{(\omega_p^2 - \omega_s^2)((\omega_p \alpha)^2 - (\omega_p^2 - \omega^2)(\omega_s^2 - \omega^2))}{((\omega_p^2 - \omega^2)^2 + (\omega_p \alpha)^2)^2}$$

$$\frac{d|Z|}{d\omega} = \frac{2|Z|}{\omega(\omega_p C_0)^2} \frac{(\omega_p^2 - \omega_s^2)((\omega_p \alpha)^2 - (\omega_p^2 - \omega^2)(\omega_s^2 - \omega^2))}{((\omega_p^2 - \omega^2)^2 + (\omega_p \alpha)^2)^2}$$

Let us find the frequencies at which the derivative $|Z|$ is reset to zero:

$$(\omega_p \alpha)^2 - (\omega_p^2 - \omega^2)(\omega_s^2 - \omega^2) = 0$$

$$-\omega^4 + \omega^2(\omega_p^2 + \omega_s^2) + (\omega_p \alpha)^2 - (\omega_p \omega_s)^2 = 0.$$

Solution of the quadratic equation:

$$\omega = \sqrt{\frac{(\omega_s^2 + \omega_p^2) \pm \sqrt{(\omega_s^2 + \omega_p^2)^2 + 4\omega_p^2(\alpha^2 - \omega_s^2)}}{2}}$$

Using the approximation $\alpha \ll \omega_s$, we obtain the frequencies at which the derivative $|Z|$ is reset: $\omega = \omega_s$ and $\omega = \omega_p$.

Note. In paragraph A12 we will show the validity of this neglect for a given quartz resonator.

Now let's determine which of the two points has resonance and which has antiresonance:

$$|Z|^2(\omega_p) = \frac{1}{(\omega_p C_0)^2} \cdot \frac{(\omega_s^2 - \omega_p^2)^2 + (\omega_p \alpha)^2}{(\omega_p \alpha)^2}$$

$$|Z|^2(\omega_s) = \frac{1}{(\omega_p C_0)^2} \cdot \frac{(\omega_p \alpha)^2}{(\omega_s^2 - \omega_p^2)^2 + (\omega_p \alpha)^2}$$

Now we can see that at $\omega = \omega_s$ there will be antiresonance, and at $\omega = \omega_p$ there will be resonance.

Answer: Answer:

$$f_{res} = \frac{\omega_p}{2\pi}; \quad f = \frac{\omega_s}{2\pi}$$

A9

0.45 pts

Remove the dependence of the resonance frequencies f_{res} and antiresonance f of the quartz resonator on the external capacitance C_{ext} . As an external capacitance C_{ext} , it is enough to use each of the provided capacitors separately, without combining them.

Answer: Answer: C_{ext} , pF f_{res} , MHz f , MHz 0 1.999850 1.999750 6.0 1.999842 1.999755 15 1.999832 1.999755 10 1.999837 1.999755 27 1.999822 1.999755 33 1.999816 1.999755 47 1.999807 1.999755 56 1.999804 1.999755 68 1.999798 1.999755 82 1.999794 1.999755

A10

0.80 pts

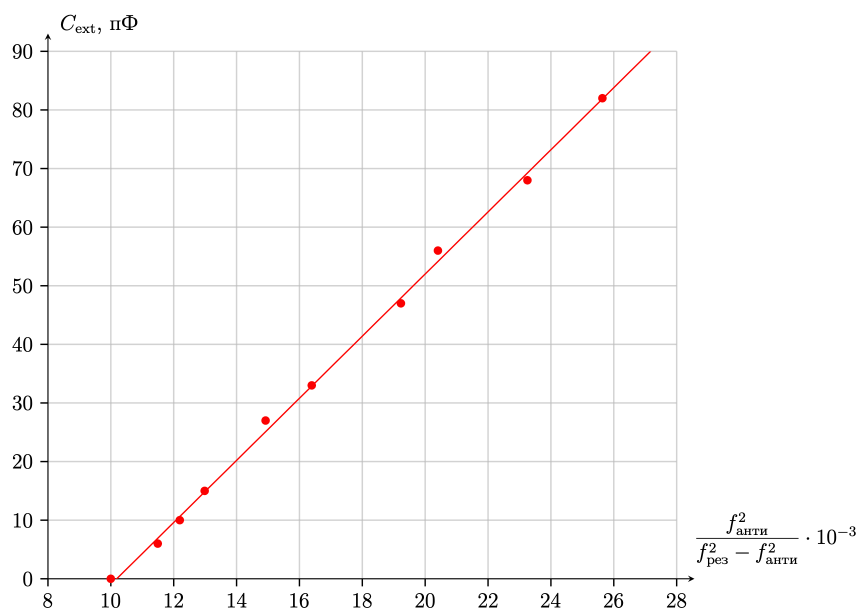
Construct a linearized graph of C_{ext} versus f_{res} and f .

Let us write the expression for the resonant frequency of a quartz resonator with a parallel connection C_{ext} :

$$\omega_p(C_{ext}) = \sqrt{\frac{1}{LC} \left(1 + \frac{C}{C_0 + C_{ext}} \right)} = \omega_s \sqrt{1 + \frac{C}{C_0 + C_{ext}}}$$

Then after the transformation we can obtain linearization:

$$C_{ext} = \frac{f^2 C}{f_{res}^2 - f^2} - C_0$$



A11

1.30 pts

From the results of points A6, A10, obtain the values L, C, C_0 .

From the linearized graph in A10 we obtain the slope and intercept, which are equal to:

$$C = 5.3 ;$$

$$C_0 = 53.9 \text{ pF}.$$

Now, knowing the value of $f = 1.999740 \text{ MHz}$, we can obtain the value of L :

$$L = \frac{1}{4\pi^2 f_{\text{анти}}^2 \cdot C} = 1.20 \text{ H}.$$

Now, knowing the value of $f = 1.999740 \text{ MHz}$, we can get the value of L :

$$L = \frac{1}{4\pi^2 f_{\text{анти}}^2 \cdot C} = 1.20 \text{ H}.$$

Answer: Answer:

$$L = 1.20 \text{ H};$$

$$C = 5.3 ;$$

$$C_0 = 53.9 \text{ pF}.$$

A12

1.20 pts

From the results of points A4, A6, A10, get the value R .

Let us first obtain the answer with the approximation $\frac{4\omega_p^2 \alpha^2}{(\omega_p^2 - \omega_s^2)^2} \ll 1$, which follows from the condition by simply expressing α from the formula obtained in paragraph A8. In this case $\alpha = 164.4 \frac{1}{\text{H}}$.

Note. It is possible to find R without approximation from the condition (this was not required from the participants). In paragraph A6, we received that $\alpha = \frac{R}{L}$, remains to be determined α . The easiest way to determine it is by resonance Z . As has already been

found:

$$\omega = \sqrt{\frac{(\omega_s^2 + \omega_p^2) \pm \sqrt{(\omega_s^2 + \omega_p^2)^2 + 4\omega_p^2(\alpha^2 - \omega_s^2)}}{2}}$$

for extrema $|Z|$. Let's substitute them into the expression for $|Z|^2$:

$$|Z|_{\text{crit}}^2 = \frac{\left(\omega_s^2 - \frac{(\omega_s^2 + \omega_p^2) \pm \sqrt{(\omega_s^2 + \omega_p^2)^2 + 4\omega_p^2(\alpha^2 - \omega_s^2)}}{2}\right)^2 + (\omega_p \alpha)^2}{(\omega_p C_0)^2 \left(\left(\omega_p^2 - \frac{(\omega_s^2 + \omega_p^2) \pm \sqrt{(\omega_s^2 + \omega_p^2)^2 + 4\omega_p^2(\alpha^2 - \omega_s^2)}}{2}\right)^2 + (\omega_p \alpha)^2\right)},$$

here sign + correspond $|Z|_{\text{max}}^2$, sign - correspond $|Z|_{\text{min}}^2$. Transform given expression:

$$|Z|_{\text{crit}}^2 = \frac{\left(\omega_p^2 - \omega_s^2 \pm \sqrt{(\omega_p^2 - \omega_s^2)^2 + 4\omega_p^2 \alpha^2}\right)^2 + 4\omega_p^2 \alpha^2}{\left(\omega_p^2 - \omega_s^2 \mp \sqrt{(\omega_p^2 - \omega_s^2)^2 + 4\omega_p^2 \alpha^2}\right)^2 + 4\omega_p^2 \alpha^2} \cdot \frac{1}{\omega_p^2 C_0^2},$$

$$|Z|_{\text{crit}}^2 = \frac{\left(1 \pm \sqrt{1 + \frac{4\omega_p^2 \alpha^2}{(\omega_p^2 - \omega_s^2)^2}}\right)^2 + \frac{4\omega_p^2 \alpha^2}{(\omega_p^2 - \omega_s^2)^2}}{\left(1 \mp \sqrt{1 + \frac{4\omega_p^2 \alpha^2}{(\omega_p^2 - \omega_s^2)^2}}\right)^2 + \frac{4\omega_p^2 \alpha^2}{(\omega_p^2 - \omega_s^2)^2}} \cdot \frac{1}{\omega_p^2 C_0^2}.$$

Introduce notation $y = \frac{4\omega_p^2 \alpha^2}{(\omega_p^2 - \omega_s^2)^2}$:

$$|Z|_{\text{crit}}^2 = \frac{(1 \pm \sqrt{1+y})^2 + y}{(1 \mp \sqrt{1+y})^2 + y} \cdot \frac{1}{\omega_p^2 C_0^2}.$$

Determine $|Z|_{\text{max}}$ we гораздо more precisely, чем $|Z|_{\text{min}}$, therefore by $|Z|_{\text{max}} \approx 12500$ Ohm, find by graph in part A4 find y :

$$|Z|_{\text{max}}^2 \cdot \omega_p^2 \cdot C_0^2 = 71.67 = \frac{(\sqrt{1+y} + 1)^2 + y}{(\sqrt{1+y} - 1)^2 + y}.$$

Solve given equation for means numerical method (method итераций, table in калькуляторе or any other рабочий method):

$$y = 0.05740 \Rightarrow \alpha = 165.6 \Rightarrow R = L \cdot \alpha = 199 \Omega,$$

which is quite close to the approximate answer obtained above, that is, the approximation from the condition for a given quartz resonator is correct.

Answer: Answer:

$$R = 197 \Omega.$$